Procedure number	PY SMM/vl/em/002	

Critical Systems

PY SMM/vl/em/002 – Critical Systems

1. Overview

1.1 The company is required to establish procedures in its SMS to identify equipment and technical systems; the sudden failure of which may result in hazardous situations. The SMS should provide for specific measures aimed at promoting the reliability of such equipment or systems. These measures should include the testing of stand-by arrangements and equipment or technical systems that are not in continuous use, in accordance with manufacturers or regulatory requirements, the identification of and carriage of adequate spare parts in support of continuous operation of the safety critical equipment. Consideration should be made to preventing cyber-attacks on critical systems.

2. Scope

2.1 This procedure applies to all safety critical equipment shown below, but is not limited to:

- All LSA and FFE inclusive of Rescue boat (and hatch)
- Fire Suppression system (Water mist etc, fire main, hydrants etc)
- Emergency shut downs (Fuel, vent closing, flaps / dampers etc)
- Fire / Smoke detection system.
- Sliding fire doors
- Shell doors

- Vapour / Gas detectors
- Steering gear
- Sewage plant
- Fuel oil system integrity (filters, centrifuge, sampling etc)
- Ground tackle (Anchors, cables, windlass´s)
- All lifting equipment (cranes, shackles, falls etc)
- Navigation equipment
- GMDSS equipment and batteries etc
- Emergency generator
- Oily Water Separator
- Potable water filtration / treatment plants
- Alarm and monitoring system
- Passenger Lift
- PA System
- SASS

2.2 A matrix is below provided to help determine criticality.

3. References

- *SMM/vl/main/001 – Equipment Maintenance*
- *SMM/vl/admin/005 – Record Keeping*

4. Critical Equipment Risk Matrix

Severity	***Likelihood***				
	1. V. Unlikely	2. Unlikely	*3. Possible*	*4.Likely*	*5. Very Likely*
	Very low never heard of	*Low Several Years Interval*	*Medium Two Years Interval*	*High Yearly Interval*	*Very High Weekly - Daily*

01 Negligible	*Environment = < 10L* *Personnel = No treatment* *Damage = None*	1	2	3	4	5
02 Slightly Harmful	*Environment = 10L -50L* *Personnel = First Aid* *Damage = Minor*	6	7	8	9	10
03 Moderate	*Environment = 50L – 500L* *Personel=Medical Treatment* *Damage = Moderate*	11	12	13	14	15
04 Harmful	*Environment = 500L-5000L* *Personel=Moderate Injury* *Damage= Severe*	16	17	18	19	20
05 Extremely Harmful	*Environment= > 5000L* *Personnel=severe injury – death* *Damage= loss of Yacht*	21	22	23	24	25

Criticality	***Description***
Very High	**Unacceptable – High risk of hazardous events** **= Extremely Hazardous to either life, the environment or property.**
Critical	**Criticality to be documented and addressed in PMS** **= Critical**
Medium	**Acceptable risk based on existing safeguards (I.E. redundancy, regular testing, maintenance, availability of spares) = Not Critical**

Low	**Low risk of hazardous events related to failure** **= Not Critical**

5. Procedure – Critical Systems